

JEE Mains 2019 Chapter wise Question Bank

Vector Algebra – Questions

Q1

Let $\vec{a} = \hat{i} - \hat{j}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that

$\vec{a} \times \vec{c} + \vec{b} = \vec{0}$ and $\vec{a} \cdot \vec{c} = 4$, then $|\vec{c}|^2$ is equal to:

- (1) $\frac{19}{2}$ (2) 9
(3) 8 (4) $\frac{17}{2}$

9 Jan Morning

Q2

Let $\vec{a} = \hat{i} + \hat{j} + \sqrt{2}\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + \sqrt{2}\hat{k}$ and $\vec{c} = 5\hat{i} + \hat{j} + \sqrt{2}\hat{k}$ be three vectors such that the projection vector of $\vec{b}$ on $\vec{a}$ is $\vec{a}$.

If $\vec{a} + \vec{b}$ is perpendicular to $\vec{c}$, then $|\vec{b}|$ is equal to:

- (1) $\sqrt{32}$ (2) 6
(3) $\sqrt{22}$ (4) 4

9 Jan Evening

Q3

Let $\vec{a} = 2\hat{i} + \lambda_1\hat{j} + 3\hat{k}$, $\vec{b} = 4\hat{i} + (3 - \lambda_2)\hat{j} + 6\hat{k}$ and

$\vec{c} = 3\hat{i} + 6\hat{j} + (\lambda_3 - 1)\hat{k}$ be three vectors such that $\vec{b} = 2\vec{a}$ and $\vec{a}$ is perpendicular to $\vec{c}$. Then a possible value of $(\lambda_1, \lambda_2, \lambda_3)$ is:

- (1) (1, 3, 1) (2) $(-\frac{1}{2}, 4, 0)$
(3) $(\frac{1}{2}, 4, -2)$ (4) (1, 5, 1)

10 Jan Morning

Q4

Let $\vec{\alpha} = (\lambda - 2)\vec{a} + \vec{b}$ and $\vec{\beta} = (4\lambda - 2)\vec{a} + 3\vec{b}$ be two given vectors where vectors $\vec{a}$ and $\vec{b}$ are non-collinear.

The value of λ for which vectors $\vec{\alpha}$ and $\vec{\beta}$ are collinear, is:

- (1) -4 (2) -3
(3) 4 (4) 3

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Q5

Let $\vec{a} = \hat{i} + 2\hat{j} + 4\hat{k}$, $\vec{b} = \hat{i} + \lambda\hat{j} + 4\hat{k}$ and $\vec{c} = 2\hat{i} + 4\hat{j} + (\lambda^2 - 1)\hat{k}$ be coplanar vectors. Then the non-zero vector $\vec{a} \times \vec{c}$ is:

- (1) $-10\hat{i} - 5\hat{j}$ (2) $-14\hat{i} - 5\hat{j}$
(3) $-14\hat{i} + 5\hat{j}$ (4) $-10\hat{i} + 5\hat{j}$

11 Jan Morning

Q6

Let $\sqrt{3}\hat{i} + \hat{j}$, $\hat{i} + \sqrt{3}\hat{j}$ and $\beta\hat{i} + (1 - \beta)\hat{j}$ respectively be the position vectors of the points A, B and C with respect to the origin O. If the distance of C from the bisector of the acute

angle between OA and OB is $\frac{3}{\sqrt{2}}$, then the sum of all

possible values of β is:

- (1) 4 (2) 3
(3) 2 (4) 1

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Q7

Vector Algebra

The sum of the distinct real values of μ , for which the vectors, $\mu\hat{i} + \hat{j} + \hat{k}$, $\hat{i} + \mu\hat{j} + \hat{k}$, $\hat{i} + \hat{j} + \mu\hat{k}$, are co-planar, is:

- (1) -1 (2) 0
(3) 1 (4) 2

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Q8

Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three unit vectors, out of which vectors $\vec{b}$ and $\vec{c}$ are non-parallel. If α and β are the angles which vector $\vec{a}$ makes with vectors $\vec{b}$ and $\vec{c}$ respectively and

$\vec{a} \times (\vec{b} \times \vec{c}) = \frac{1}{2}\vec{b}$, then $|\alpha - \beta|$ is equal to:

- (1) 30° (2) 90°
(3) 60° (4) 45°

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Q9

The magnitude of the projection of the vector $2\hat{i} + 3\hat{j} + \hat{k}$ on the vector perpendicular to the plane containing the vectors $\hat{i} + \hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} + 3\hat{k}$, is:

- (1) $\frac{\sqrt{3}}{2}$ (2) $\sqrt{6}$ (3) $3\sqrt{6}$ (4) $\sqrt{\frac{3}{2}}$

8 April Morning

Q10

Let $\vec{a} = 3\hat{i} + 2\hat{j} + x\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, for some real x .

Then $|\vec{a} \times \vec{b}| = r$ is possible if:

- (1) $\sqrt{\frac{3}{2}} < r \leq 3\sqrt{\frac{3}{2}}$ (2) $r \geq 5\sqrt{\frac{3}{2}}$
(3) $0 < r \leq \sqrt{\frac{3}{2}}$ (4) $3\sqrt{\frac{3}{2}} < r < 5\sqrt{\frac{3}{2}}$

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Q11

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Let $\vec{\alpha} = 3\hat{i} + \hat{j}$ and $\vec{\beta} = 2\hat{i} - \hat{j} + 3\hat{k}$. If $\vec{\beta} = \vec{\beta}_1 - \vec{\beta}_2$, where $\vec{\beta}_1$ is parallel to $\vec{\alpha}$ and $\vec{\beta}_2$ is perpendicular to $\vec{\alpha}$, then $\vec{\beta}_1 \times \vec{\beta}_2$ is equal to:

- (1) $-3\hat{i} + 9\hat{j} + 5\hat{k}$ (2) $3\hat{i} - 9\hat{j} - 5\hat{k}$
(3) $\frac{1}{2}(-3\hat{i} + 9\hat{j} + 5\hat{k})$ (4) $\frac{1}{2}(3\hat{i} - 9\hat{j} + 5\hat{k})$

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Q12

If a unit vector $\vec{a}$ makes angles $\pi/3$ with $\hat{i}$, $\pi/4$ with $\hat{j}$ and $\theta \in (0, \pi)$ with $\hat{k}$, then a value of θ is:

- (1) $\frac{5\pi}{6}$ (2) $\frac{\pi}{4}$ (3) $\frac{5\pi}{12}$ (4) $\frac{2\pi}{3}$

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Q13

The distance of the point having position vector $-\hat{i} + 2\hat{j} + 6\hat{k}$ from the straight line passing through the point $(2, 3, -4)$ and parallel to the vector, $6\hat{i} + 3\hat{j} - 4\hat{k}$ is:

- (1) 7 (2) $4\sqrt{3}$ (3) 6 (4) $2\sqrt{13}$

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Q14

Let $a = 3i + 2j + 2k$ and $b = i + 2j - 2k$ be two vectors. If a vector perpendicular to both the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ has the magnitude 12 then one such vector is:

- (1) $4(2\hat{i} + 2\hat{j} + 2\hat{k})$ (2) $4(2\hat{i} - 2\hat{j} - \hat{k})$
(3) $4(2\hat{i} + 2\hat{j} - \hat{k})$ (4) $4(-2\hat{i} - 2\hat{j} + \hat{k})$

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Q15

Let $\alpha \in \mathbb{R}$ and the three vectors $\vec{a} = \alpha \hat{i} + \hat{j} + 3\hat{k}$,

$\vec{b} = 2\hat{i} + \hat{j} - \alpha \hat{k}$ and $\vec{c} = \alpha \hat{i} - 2\hat{j} + 3\hat{k}$. Then the set

$S = \{\alpha : \vec{a}, \vec{b} \text{ and } \vec{c} \text{ are coplanar}\}$

- (1) is singleton
- (2) is empty
- (3) contains exactly two positive numbers
- (4) contains exactly two numbers only one of which is positive

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Vector Algebra - Solutions

Q1

$$(1) \because |\vec{a} \times \vec{c}|^2 = |\vec{a}|^2 |\vec{c}|^2 - (\vec{a} \cdot \vec{c})^2$$

$$\Rightarrow |-\vec{b}|^2 = 2|\vec{c}|^2 - 16$$

$$\Rightarrow 3 = 2|\vec{c}|^2 - 16$$

$$\Rightarrow |\vec{c}|^2 = \frac{19}{2}$$

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Q2

$$(2) \text{ Projection of } \vec{b} \text{ on } \vec{a} = \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|} = \frac{b_1 + b_2 + 2}{4}$$

$$\text{According to question } \frac{b_1 + b_2 + 2}{2} = \sqrt{1+1+2} = 2$$

$$\Rightarrow b_1 + b_2 = 2 \quad \dots(1)$$

Since, $\vec{a} + \vec{b}$ is perpendicular to $\vec{c}$.

$$\text{Hence, } \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c} = 0$$

$$\Rightarrow 8 + 5b_1 + b_2 + 2 = 0 \quad \dots(2)$$

From (1) and (2),

$$b_1 = -3, b_2 = 5$$

$$\Rightarrow \vec{b} = -3\hat{i} + 5\hat{j} + \sqrt{2}\hat{k}$$

$$|\vec{b}| = \sqrt{9+25+2} = 6$$

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Q3

$$(2) \because \vec{b} = 2\vec{a}$$

$$\therefore 4\hat{i} + (3 - \lambda_2)\hat{j} + 6\hat{k} = 4\hat{i} + 2\lambda_1\hat{j} + 6\hat{k}$$

$$\therefore 3 - \lambda_2 = 2\lambda_1 \quad \dots(1)$$

$\therefore \vec{a}$ is perpendicular to $\vec{c}$

$$\therefore \vec{a} \cdot \vec{c} = 0$$

$$\Rightarrow 6 + 6\lambda_1 + 3(\lambda_3 - 1) = 0$$

$$\Rightarrow 2 + 2\lambda_1 + \lambda_3 - 1 = 0$$

$$\Rightarrow \lambda_3 = -2\lambda_1 - 1 \quad \dots(2)$$

Since $\left(\frac{-1}{2}, 4, 0\right)$ satisfies equation (1) and (2). Hence, one of possible value of

$$\lambda_1 = -\frac{1}{2}, \lambda_2 = 4 \text{ and } \lambda_3 = 0$$

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Q4

Vector Algebra

(1) Let $\vec{\alpha}$ and $\vec{\beta}$ are collinear for same k

i.e., $\vec{\alpha} = k\vec{\beta}$

$$(\lambda - 2)\vec{a} + \vec{b} = k((4\lambda - 2)\vec{a} + 3\vec{b})$$

$$(\lambda - 2)\vec{a} + \vec{b} = k(4\lambda - 2)\vec{a} + 3k\vec{b}$$

$$(\lambda - 2 - k(4\lambda - 2))\vec{a} + \vec{b}(1 - 3k) = 0$$

But $\vec{a}$ and $\vec{b}$ are non-collinear, then

$$\lambda - 2 - k(4\lambda - 2) = 0, 1 - 3k = 0$$

$$\Rightarrow k = \frac{1}{3} \text{ and } \lambda - 2 - \frac{1}{3}(4\lambda - 2) = 0$$

$$3\lambda - 6 - 4\lambda + 2 = 0$$

$$\lambda = -4$$

Q5

(4) $\therefore \vec{a}, \vec{b}$ and $\vec{c}$ are coplanar

$$\therefore \begin{vmatrix} 1 & 2 & 4 \\ 1 & \lambda & 4 \\ 2 & 4 & (\lambda^2 - 1) \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - \lambda - 16 + 2(8 - \lambda^2 + 1) + 4(4 - 2\lambda) = 0$$

$$\Rightarrow \lambda^3 - 2\lambda^2 - 9\lambda + 18 = 0$$

$$\text{i.e., } (\lambda - 2)(\lambda - 3)(\lambda + 3) = 0$$

$$\text{For } \lambda = 2, \vec{c} = 2\hat{i} + 4\hat{j} + 3\hat{k}$$

$$\therefore \vec{a} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 4 \\ 2 & 4 & 3 \end{vmatrix} = -10\hat{i} + 5\hat{j}$$

$$\text{For } \lambda = 3 \text{ or } -3, \vec{c} = 2\vec{a} \Rightarrow \vec{a} \times \vec{c} = 0 \quad (\text{Rejected})$$

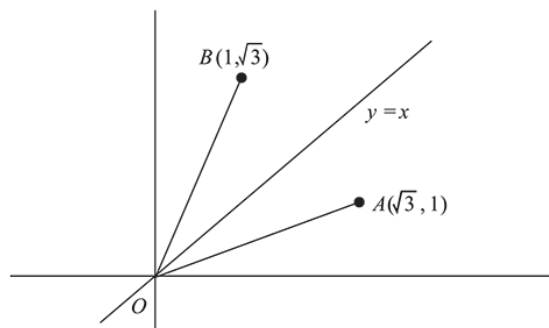
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Q6

(4) Since, the angle bisector of acute angle between OA and OB would be $y = x$



Since, the distance of C from bisector $= \frac{3}{\sqrt{2}}$

$$\Rightarrow \left| \frac{\beta - (1 - \beta)}{\sqrt{2}} \right| = \frac{3}{\sqrt{2}} = 2\beta = \pm 3 + 1$$

$$\beta = 2 \text{ or } \beta = -1$$

Hence, the sum of all possible value of $\beta = 2 + (-1) = 1$

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Q7

(1) $\therefore$ Three vectors $(\mu\hat{i} + \hat{j} + \hat{k}), (\hat{i} + \mu\hat{j} + \hat{k})$ and $(\hat{i} + \hat{j} + \mu\hat{k})$ are coplanar.

$$\therefore \begin{vmatrix} \mu & 1 & 1 \\ 1 & \mu & 1 \\ 1 & 1 & \mu \end{vmatrix} = 0$$

$$\Rightarrow \mu(\mu^2 - 1) + 1 - \mu + 1 - \mu = 0$$

$$\Rightarrow (1 - \mu)[2 - \mu(\mu + 1)] = 0$$

$$\Rightarrow (1 - \mu)[\mu^2 + \mu - 2] = 0$$

$$\Rightarrow \mu = 1, -2$$

Therefore, sum of all real values $= 1 - 2 = -1$

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Q8

Vector Algebra

(1) Since, $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three unit vectors

$$\therefore |\vec{a}| = |\vec{b}| = |\vec{c}| = 1$$

$$\text{Then, } \vec{a} \times (\vec{b} \times \vec{c}) = \frac{1}{2} \vec{b}$$

$$(\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} = \frac{1}{2} \vec{b}$$

$$\therefore \vec{a} \cdot \vec{c} = \frac{1}{2} \text{ and } \vec{a} \cdot \vec{b} = 0$$

$$|\vec{a}| |\vec{c}| \cos \beta = \frac{1}{2} \text{ and } |\vec{a}| |\vec{b}| \cos \alpha = 0$$

$$\Rightarrow \beta = 60^\circ \text{ and } \alpha = 90^\circ$$

$$\text{Hence, } |\alpha - \beta| = |90^\circ - 60^\circ| = 30^\circ$$

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Q9

(4) Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$

$\therefore$ Vector perpendicular to $\vec{a}$ and $\vec{b}$ is $\vec{a} \times \vec{b}$

$$\therefore \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & 2 & 3 \end{vmatrix} = \hat{i} - 2\hat{j} + \hat{k}$$

Now, projection of vector $\vec{c} = 2\hat{i} + 3\hat{j} + \hat{k}$ on $\vec{a} \times \vec{b}$ is

$$= \frac{|\vec{c} \cdot (\vec{a} \times \vec{b})|}{|\vec{a} \times \vec{b}|} = \frac{|2 - 6 + 1|}{\sqrt{6}}$$

$$= \frac{3}{\sqrt{6}} = \sqrt{\frac{3}{2}}$$

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Q10

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(2) Given, $\vec{a} = 3\hat{i} + 2\hat{j} + x\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$

$$\text{Now, } \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & x \\ 1 & -1 & 1 \end{vmatrix}$$

$$= (2+x)\hat{i} + (x-3)\hat{j} - 5\hat{k}$$

$$|\vec{a} \times \vec{b}| = \sqrt{(2+x)^2 + (x-3)^2 + (-5)^2} = r$$

$$\Rightarrow r = \sqrt{4+x^2+4x+x^2+9-6x+25}$$

$$= \sqrt{2x^2-2x+38} = \sqrt{2\left(x^2-x+\frac{1}{4}\right)+38-\frac{1}{2}}$$

$$= \sqrt{2\left(x-\frac{1}{2}\right)^2 + \frac{75}{2}} \Rightarrow r \geq \sqrt{\frac{75}{2}} \Rightarrow r \geq 5\sqrt{\frac{3}{2}}$$

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Q11

Vector Algebra

$$(3) \vec{\beta} = \vec{\beta}_1 - \vec{\beta}_2 \quad \dots(1)$$

Since, $\vec{\beta}_2$ is perpendicular to $\vec{\alpha}$.

$$\therefore \vec{\beta}_2 \cdot \vec{\alpha} = 0$$

Since, $\vec{\beta}_1$ is parallel to $\vec{a}$.

then $\vec{\beta}_1 = \lambda \vec{\alpha}$ (say)

$$\vec{a} \cdot \vec{\beta} = \vec{a} \cdot \vec{\beta}_1 - \vec{a} \cdot \vec{\beta}_2$$

$$\Rightarrow 5 = \lambda \alpha^2 \Rightarrow 5 = \lambda \times 10 \quad (\because |\vec{\alpha}| = \sqrt{10}).$$

$$\Rightarrow \lambda = \frac{1}{2} \quad \therefore \vec{\beta}_1 = \frac{\vec{\alpha}}{2}$$

$$\therefore \vec{\beta}_1 = \frac{\vec{\alpha}}{2}$$

Cross product with $\vec{\beta}_1$ in equation (1)

$$\Rightarrow \vec{\beta} \times \vec{\beta}_1 = -\vec{\beta}_2 \times \vec{\beta}_1$$

$$\Rightarrow \vec{\beta} \times \vec{\beta}_1 = \vec{\beta}_1 \times \vec{\beta}_2 \Rightarrow \vec{\beta}_1 \times \vec{\beta}_2 = \frac{-\vec{\beta} \times \vec{\alpha}}{2}$$

$$\Rightarrow \vec{\beta}_1 \times \vec{\beta}_2 = \frac{1}{2} \begin{vmatrix} i & j & k \\ 2 & -1 & 3 \\ 3 & 1 & 0 \end{vmatrix}$$

$$= \frac{1}{2} [-3\hat{i} - \hat{j}(-9) + \hat{k}(5)]$$

$$= \frac{1}{2} [-3\hat{i} + 9\hat{j} + 5\hat{k}]$$

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Q12

(4) Let $\cos \alpha, \cos \beta, \cos \gamma$ be direction cosines of $\vec{a}$.

$$\therefore \cos \alpha = \cos \frac{\pi}{3}, \cos \beta = \cos \frac{\pi}{4} \text{ and } \cos \gamma = \cos \theta$$

$$\Rightarrow \cos^2 \frac{\pi}{3} + \cos^2 \frac{\pi}{4} + \cos^2 \theta = 1 \Rightarrow \cos^2 \theta = \frac{1}{4}$$

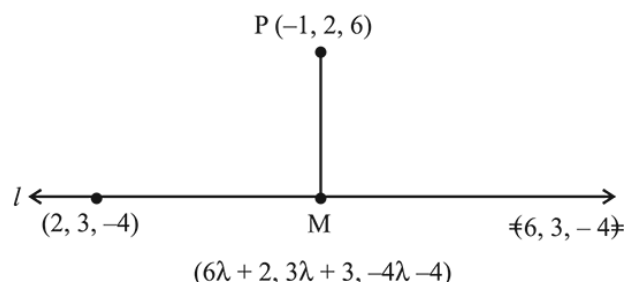
$$\Rightarrow \cos \theta = \pm \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3} \text{ or } \frac{2\pi}{3}$$

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Q13

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$$(1) \text{ Equation of the line is } \frac{x-2}{6} = \frac{y-3}{3} = \frac{z+4}{-4}$$



Let the point M on the line l is $(6\lambda + 2, 3\lambda + 3, -4\lambda - 4)$

Direction ratio's of PM is $(6\lambda + 3, 3\lambda + 1, -4\lambda - 10)$

$$\Rightarrow (6\lambda + 3)(6) + (3\lambda + 1)(3) + (-4\lambda - 10)(-4) = 0$$

$$\Rightarrow \lambda = -1 \Rightarrow M \equiv (-4, 0, 0)$$

$$\therefore PM = \sqrt{9+4+36} = 7$$

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Q14

(2) Let vector be $\lambda[(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})]$

Given, $\vec{a} = 3\hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}$

$$\therefore \vec{a} + \vec{b} = 4\hat{i} + 4\hat{j} \text{ and } \vec{a} - \vec{b} = 2\hat{i} + 4\hat{k}$$

$$\therefore \text{vector} = \lambda[(4\hat{i} + 4\hat{j}) \times (2\hat{i} + 4\hat{k})]$$

$$= \lambda[16\hat{i} - 16\hat{j} - 8\hat{k}] = 8\lambda[2\hat{i} - 2\hat{j} - \hat{k}]$$

Given that magnitude of the vector is 12.

$$\therefore 12 = 8|\lambda| \sqrt{4+4+1} \Rightarrow |\lambda| = \frac{1}{2}$$

$$\therefore \text{required vector is } \pm 4(2\hat{i} - 2\hat{j} - \hat{k})$$

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Q15

(2) Let, three vectors $\vec{a}, \vec{b}, \vec{c}$ are coplanar, then $[\vec{a}, \vec{b}, \vec{c}] = 0$

$$\Rightarrow \begin{vmatrix} \alpha & 1 & 3 \\ 2 & 1 & -\alpha \\ \alpha & -2 & 3 \end{vmatrix} = 0 \Rightarrow \alpha^2 + 6 = 0$$

$\therefore$ no real value of ' α ' exist.

$\therefore$ set S is an empty set.

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